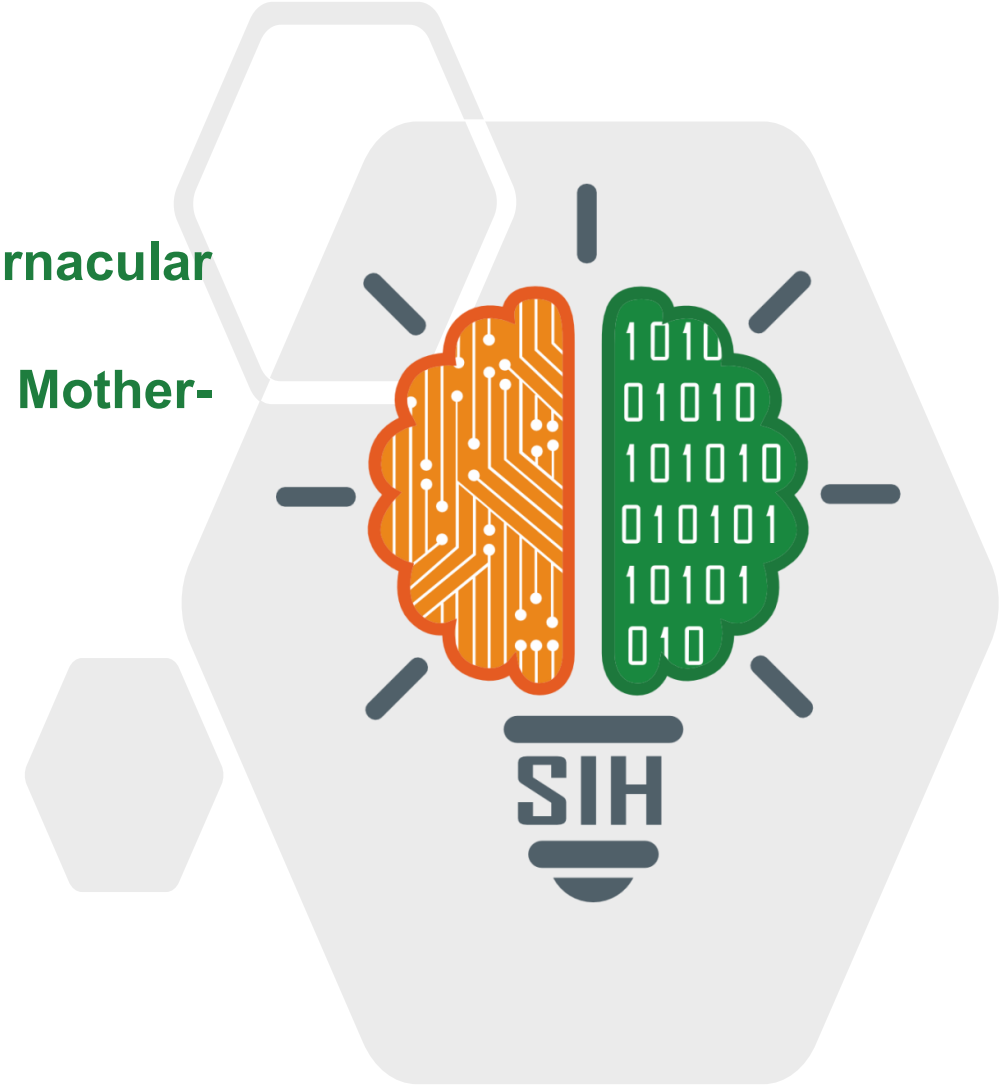
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SMART INDIA HACKATHON 2026

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- **Problem Statement ID : 26042**
- **Problem Statement Title: AI-Powered Vernacular Pedagogy &Real-Time Translation for Mother-Tongue Education**
- **Theme: Smart Education**
- **PS Category: Software**
- **Team ID:**
- **Team Name: Palash-Nova**



PALASH

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PROBLEM EXISTING

- Teacher-Language Mismatch: Students struggle to understand lessons when the teacher's language differs from their native language.
- Limited NLP Resources: Ho, Mundari, and Santhali lack sufficient digital language datasets and AI/NLP support.
- No Reliable Internet Solution: Poor connectivity makes cloud-dependent classroom tools ineffective in remote areas.

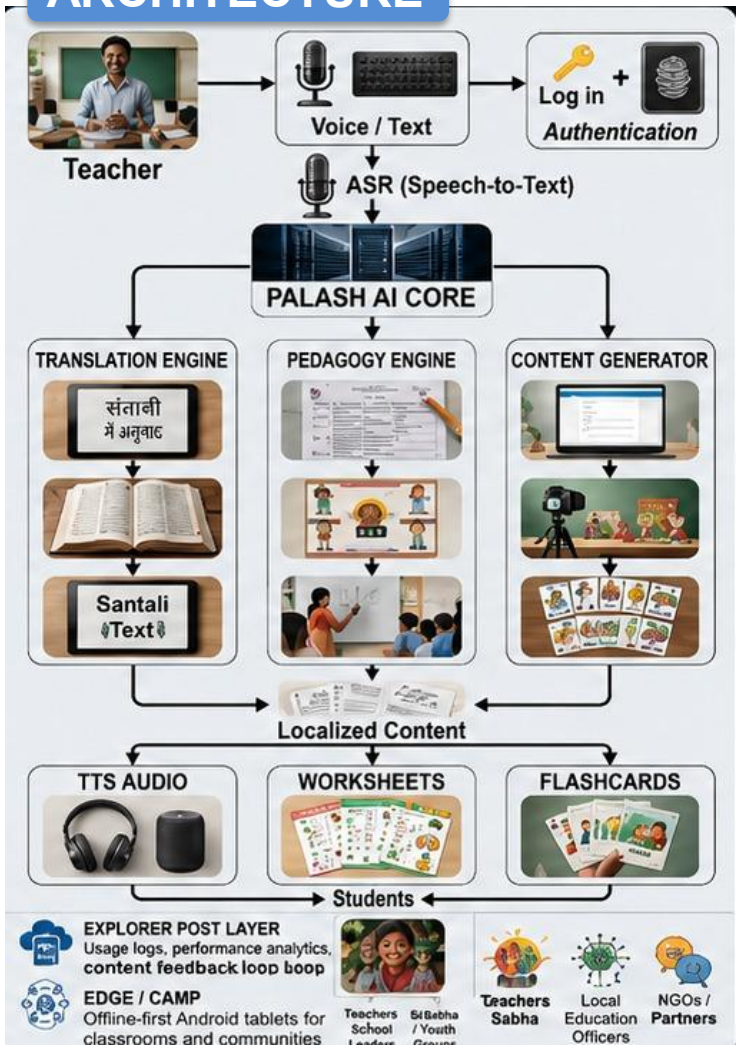
PROPOSED SOLUTION

- Offline-First Learning Assistant: Android-based teaching companion designed for low-cost tablets and remote classrooms.
- Instant Bilingual Learning: Converts Hindi content into tribal languages with both translated text and synthesized audio.
- Real-Time Voice Interaction: Enables Hindi ↔ Tribal language speech communication with a response time target of under 3 seconds.

UVP(UNIQUE VALUE PROPOSITION)

- Domain-Constrained Translation: Specialized education-focused translation designed for classroom learning instead of generic conversational AI.
- Offline-Ready Learning System: Supports offline inference and downloadable content packs, ensuring uninterrupted use after initial synchronization.
- Automated FLN Content Creation: Generates bilingual worksheets, visual flashcards, and learning materials aligned with Foundational Literacy & Numeracy (FLN) outcomes.

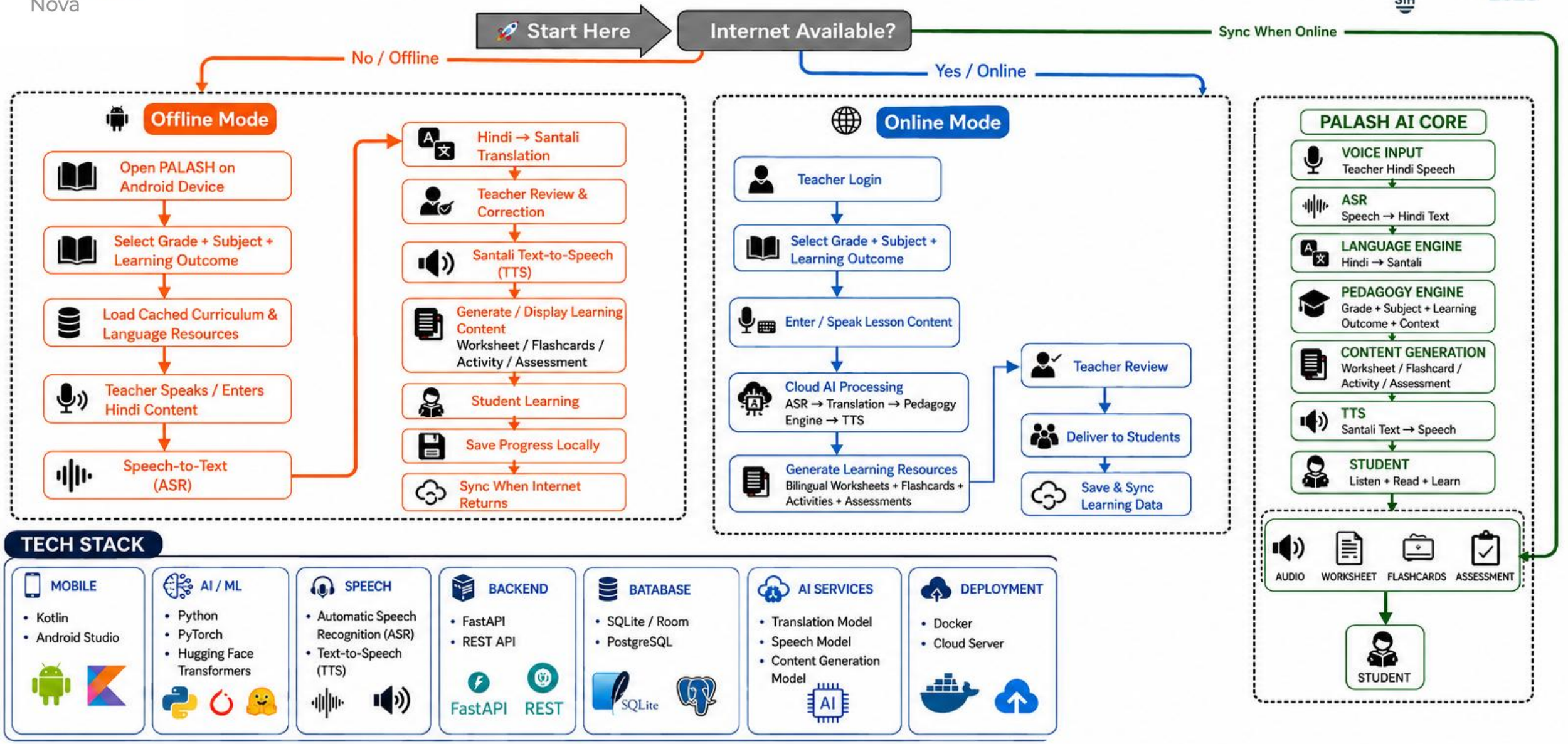
ARCHITECTURE



TECHNICAL APPROACH

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TECH STACK

MOBILE
- Kotlin
- Android Studio

AI / ML
- Python
- PyTorch
- Hugging Face Transformers

SPEECH
- Automatic Speech Recognition (ASR)
- Text-to-Speech (TTS)

BACKEND
- FastAPI
- REST API

FastAPI REST

BATABASE
- SQLite / Room
- PostgreSQL

SQLite

AI SERVICES
- Translation Model
- Speech Model
- Content Generation Model

AI

DEPLOYMENT
- Docker
- Cloud Server

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FEASIBILITY AND VIABILITY

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FEASIBILITY

- **Classroom Translation:** Hindi ↔ Tribal language text translation for educational content is technically feasible with existing NLP and MT technologies.
- **Offline Android Deployment:** Compact and quantized AI models can run on low-cost Android tablets without continuous internet access.
- **Real-Time Voice Interaction (<3s):** Achievable for a controlled classroom demo with limited vocabulary and domain-specific content, but challenging for open-ended conversations.

DATA AVAILABILITY

- **FLN Content Sources:** NIPUN Bharat and Foundational Literacy & Numeracy (FLN) materials provide publicly available educational content for alignment.
- **Open NLP Resources:** AI4Bharat offers Indic language datasets and machine translation resources, including support for Santali.
- **Speech Data Collection:** Bhashadaan can be leveraged to gather speech, transcription, and translation datasets.

VIABILITY

- **Market Need:** Millions of tribal-language students face learning barriers due to teacher-language mismatch, creating a strong demand for inclusive educational technology.
- **Policy Alignment:** Supports NIPUN Bharat, National Education Policy (NEP) 2020, Digital India,multilingual education initiatives.
- **Cost Efficiency:** Operates on affordable Android tablets & functions offline, minimizing infrastructure connectivity costs.
- **Scalability:** New tribal languages, content packs, and curriculum modules can be added incrementally without major platform changes.
- **Long-Term Impact:** Improves foundational literacy outcomes while preserving and promoting indigenous languages through digital education.

TECHNICAL CHALENEGES

- **Risk:** Limited datasets, dialect variations, device constraints, and children's speech variability affect accuracy.
- **Mitigation:** Quantized offline models, speech fine-tuning, teacher corrections, confidence checks, and dataset expansion.

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IMPACTS & BENEFITS

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- **Learning & Educational Impact:** PALASH bridges the language gap between Hindi-medium teachers and tribal-language learners by converting classroom content into understandable Santali learning resources. This can improve comprehension, participation and foundational learning outcomes.
- **Language Preservation & Inclusion:** By creating digital learning resources in Santali and supporting voice-based interaction, PALASH brings an underrepresented Indian language into the digital education ecosystem while making classrooms more inclusive.
- **Accessibility & Connectivity Gains:** PALASH uses an offline-first approach so teachers can access cached lessons, localized content and learning materials even in low-connectivity areas. Content and progress synchronize automatically when internet access returns.

FEATURE COMPARISON

Feature	Conventional Approach	PALASH
Tribal-language learning	Limited	✓
Hindi → Santali localization	Manual / Limited	✓
Voice-based learning	Limited	✓
Pedagogy-aware content	Not integrated	✓
Offline classroom support	Limited	✓
Worksheets & activities	Mostly manual	✓
Teacher review	Limited	✓
Expandable language support	Limited	✓

HOW PALASH WORKS

MODE 1: ONLINE — AI ASSISTED

- **For:** Teacher has internet connectivity and wants full AI-powered processing.
- **Process:** Teacher Input → ASR → Hindi-Santali Translation → Pedagogy Engine → Content Generation → TTS
- **Time:** Near Real-Time / Model Dependent
- **Best For:** Connected classrooms, content preparation and teacher-assisted learning.

MODE 2: OFFLINE — CLASSROOM

- **For:** Low/no internet connectivity with previously synchronized resources.
- **Process:** Open App → Select Lesson → Load Cached Content → Voice/Text Interaction → Local Processing → Learning Activity
- **Time:** Immediate for Cached Content
- **Best For:** Rural and low-connectivity classrooms.

MODE 3: SYNC — CONNECT & UPDATE

- **For:** Device reconnects to the internet after offline classroom use.
- **Process:** Reconnect → Sync Lessons & Progress → Update Language Resources → Upload Teacher Corrections → Continue Learning
- **Time:** Automatic / Background Sync
- **Best For:** Maintaining continuity between offline and online learning.

RESEARCH AND REFERENCES

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RESEARCH & TECHNICAL FOUNDATION

Low-Resource Santali NLP: Limited data, script, and morphology challenges; improved MT resources are still needed.

Santali Machine Translation: Fine-tuned IndicTrans2 outperformed ByT5, showing effective multilingual transfer learning.

Santali Speech Recognition: Whisper-based cross-lingual transfer enables Santali speech recognition in the Ol Chiki script.

Reference: ACL Anthology
https://aclanthology.org/2025.mmloso-1.14/

Reference: ACL Anthology
https://aclanthology.org/2025.mmloso-1.9/

Reference: ACL Anthology
https://aclanthology.org/2025.findings-ijcnlp.16/

EXISTING SOLUTIONS / TECHNOLOGIES

BHASHINI / Anuvaad: Government-backed multilingual text, speech, and translation platform.

IndicTrans2 & NLLB-200: Multilingual translation models supporting Indian and low-resource languages.

Whisper & MMLoSo: Speech recognition and research initiatives for low-resource tribal languages, including Santali.

WHY PALASH STANDS OUT

Education-Focused: Maps grade, subject, and learning outcomes to classroom-ready content.

Offline, Voice & Teacher-Centric: Supports offline use, voice/text input, and teacher-reviewed outputs.

Beyond Translation: Generates learning resources and supports scalable multilingual expansion.

PROJECT RESOURCES/REFERENCES

1.BHASHINI / Anuvaad
https://anuvaad.bhashini.gov.in/

2.A14Bharat / IndicTrans2
https://github.com/AI4Bharat/IndicTrans2

3. NLLB-200 (Meta)
https://ai.meta.com/blog/nllb-200-high-quality-machine-translation/

4. **MMLoSo 2025 Shared Task (Machine Translation into Tribal Languages)**
https://aclanthology.org/events/mmloso-2025/